

Distributed Systems Assignment

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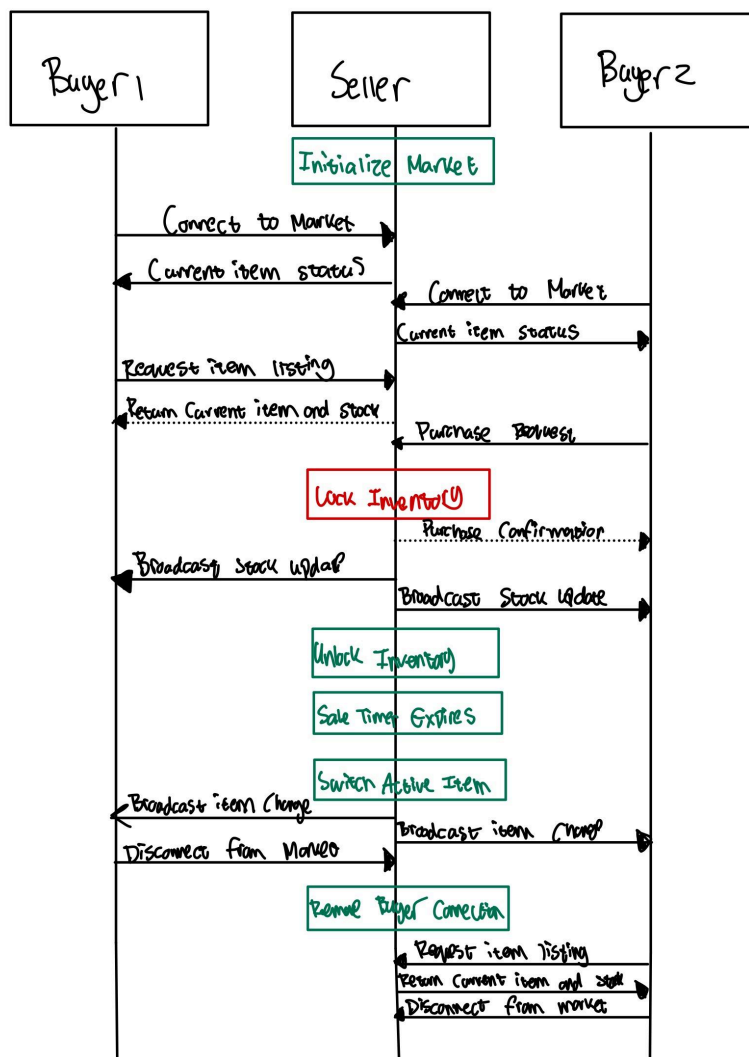
Student number: C21468162

Subject: Distributed Systems

Assignment: Distributed Systems Food Market Place

Declaration: I declare that this work, which is submitted as part of my coursework, is entirely my own, except where clearly and explicitly stated.

Applications Architecture:



Description:**Core Components:****Seller (Server):**

- Single seller acts as the central server
- Handles concurrent buyer connections through threading
- Maintains inventory state and synchronization
- Manages active item sales and timing
- Broadcasts updates to all connected buyers

Buyers (Clients):

- Multiple buyers can connect simultaneously
- Each buyer runs independently
- Maintains active connection with seller
- Receives broadcast updates
- Sends purchase/list requests

Key Architectural Features:**Client-Server Model:**

- Seller operates as server (port 5555)
- Buyers are clients connecting to server
- Centralized inventory management

Concurrency Handling:

- Multi-threaded server design
- Each buyer connection runs in separate thread

State Management:

- Centralized inventory state at seller
- Timer-based item rotation
- Connection state tracking

Error Handling:

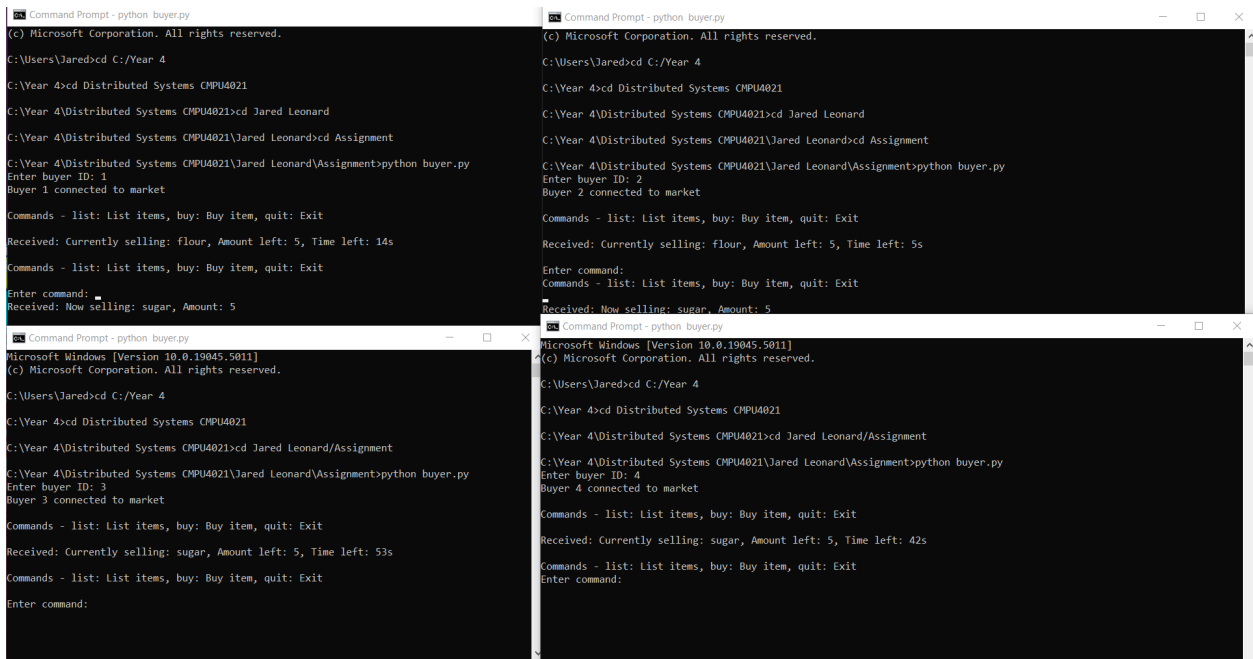
- Connection failure detection
- Client disconnection handling
- Input validation

The architecture follows a simple but robust distributed system design where all business logic and state management is centralized at the seller, while buyers act as thin clients primarily sending requests and receiving updates.

Setup and Running

Connected:

```
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python seller.py
Seller 1 started. Currently selling: flour
Waiting for buyers...
```



The image displays four terminal windows, each running the `python buyer.py` script. Each window shows the user navigating through directories to the assignment folder and then executing the script. The script prompts for a buyer ID and then displays a list of available items (flour or sugar) and their current status (amount left, time left). The user enters a command (list, buy, or quit) and the script responds accordingly.

```
Command Prompt - python buyer.py
(c) Microsoft Corporation. All rights reserved.
C:\Users\Jared>cd C:/Year 4
C:\Year 4>cd Distributed Systems CMPU4021
C:\Year 4\Distributed Systems CMPU4021>cd Jared Leonard
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard>cd Assignment
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python buyer.py
Enter buyer ID: 1
Buyer 1 connected to market
Commands - list: List items, buy: Buy item, quit: Exit
Received: Currently selling: flour, Amount left: 5, Time left: 14s
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:
Received: Now selling: sugar, Amount: 5

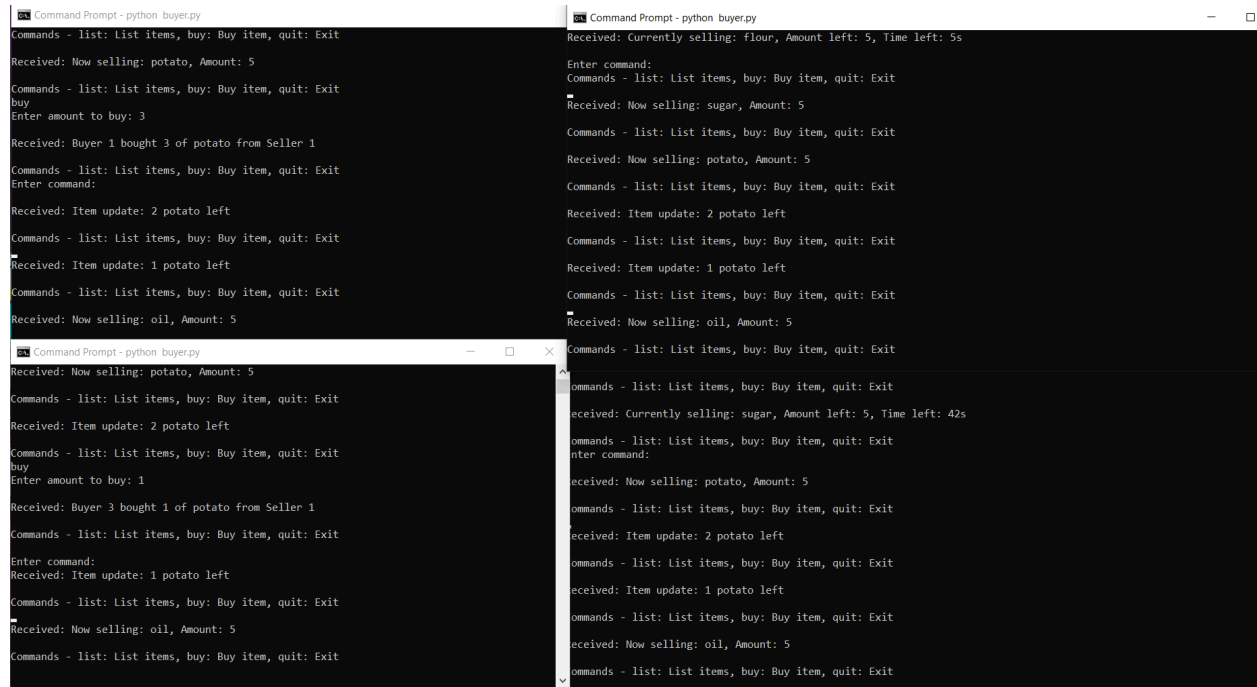
Command Prompt - python buyer.py
(c) Microsoft Corporation. All rights reserved.
C:\Users\Jared>cd C:/Year 4
C:\Year 4>cd Distributed Systems CMPU4021
C:\Year 4\Distributed Systems CMPU4021>cd Jared Leonard
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard>cd Assignment
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python buyer.py
Enter buyer ID: 2
Buyer 2 connected to market
Commands - list: List items, buy: Buy item, quit: Exit
Received: Currently selling: flour, Amount left: 5, Time left: 5s
Enter command:
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: sugar, Amount: 5

Command Prompt - python buyer.py
Microsoft Windows [Version 10.0.19045.5011]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Jared>cd C:/Year 4
C:\Year 4>cd Distributed Systems CMPU4021
C:\Year 4\Distributed Systems CMPU4021>cd Jared Leonard\Assignment
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python buyer.py
Enter buyer ID: 3
Buyer 3 connected to market
Commands - list: List items, buy: Buy item, quit: Exit
Received: Currently selling: sugar, Amount left: 5, Time left: 53s
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:

Command Prompt - python buyer.py
Microsoft Windows [Version 10.0.19045.5011]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Jared>cd C:/Year 4
C:\Year 4>cd Distributed Systems CMPU4021
C:\Year 4\Distributed Systems CMPU4021>cd Jared Leonard\Assignment
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python buyer.py
Enter buyer ID: 4
Buyer 4 connected to market
Commands - list: List items, buy: Buy item, quit: Exit
Received: Currently selling: sugar, Amount left: 5, Time left: 42s
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:
```

```
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python seller.py
Seller 1 started. Currently selling: flour
Waiting for buyers...
Buyer 1 joined the market
Buyer 2 joined the market
Buyer 3 joined the market
Buyer 4 joined the market
```

Item bought updated across all buyers



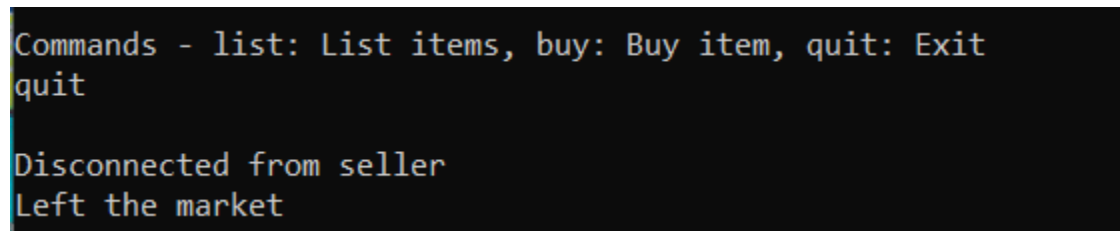
```
Command Prompt - python buyer.py
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: potato, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
buy
Enter amount to buy: 3
Received: Buyer 1 bought 3 of potato from Seller 1
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:
Received: Item update: 2 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 1 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: oil, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit

Command Prompt - python buyer.py
Received: Now selling: potato, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 2 potato left
Commands - list: List items, buy: Buy item, quit: Exit
buy
Enter amount to buy: 1
Received: Buyer 3 bought 1 of potato from Seller 1
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:
Received: Item update: 1 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: oil, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit

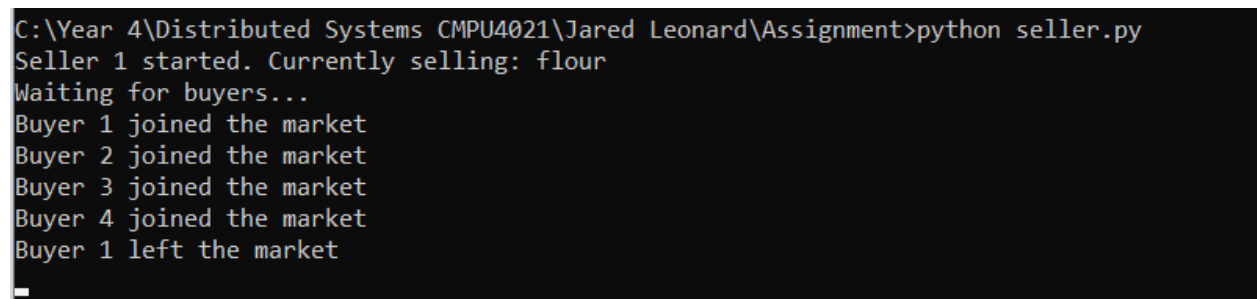
Command Prompt - python buyer.py
Received: Currently selling: flour, Amount left: 5, Time left: 5s
Enter command:
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: sugar, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: potato, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 2 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 1 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: oil, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
Received: Currently selling: sugar, Amount left: 5, Time left: 42s
Commands - list: List items, buy: Buy item, quit: Exit
Enter command:
Received: Now selling: potato, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 2 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Item update: 1 potato left
Commands - list: List items, buy: Buy item, quit: Exit
Received: Now selling: oil, Amount: 5
Commands - list: List items, buy: Buy item, quit: Exit
```

We can also see items are updated every 60 seconds

Disconnecting



```
Commands - list: List items, buy: Buy item, quit: Exit
quit
Disconnected from seller
Left the market
```



```
C:\Year 4\Distributed Systems CMPU4021\Jared Leonard\Assignment>python seller.py
Seller 1 started. Currently selling: flour
Waiting for buyers...
Buyer 1 joined the market
Buyer 2 joined the market
Buyer 3 joined the market
Buyer 4 joined the market
Buyer 1 left the market
```

Please note the directory in the command prompt

Contents of ReadMe file: How to run the system

Food Market System

A distributed system simulating an electronic marketplace where multiple buyers can purchase items from a seller.

Prerequisites

- Python installed
- Windows Operating System

System Requirements

- Each item (flour, sugar, potato, oil) starts with 5 units
- Items switch automatically after 60 seconds or when sold out
- Multiple buyers can connect simultaneously

Step 1: Setup Files

1. Save `seller.py` and `buyer.py` in the same directory
2. Open Command Prompt as administrator

Step 2: Start the Seller

1. Navigate to the directory containing the files:

```
...
```

```
cd C:/Year 4/Distributed Systems CMPU4021/Jared Leonard/Assignment
```

```
...
```

2. Start the seller:

```
...
```

```
python seller.py
```

```
...
```

You should see:

```
...
```

Seller started. Currently selling: flour

Waiting for buyers...

```
...
```

Step 3: Start Multiple Buyers

1. Open new Command Prompt windows (minimum 4 for demo)
2. In each window, navigate to the same directory:

...

```
cd C:/Year 4/Distributed Systems CMPU4021/Jared Leonard/Assignment
```

...

3. Start a buyer in each window:

...

```
python buyer.py
```

...

4. Enter a unique buyer ID when prompted in each window (e.g., buyer1, buyer2, etc.)

Step 4: Demo Commands

In any buyer window, you can use these commands:

- `list`: Show current item and stock
- `buy`: Purchase item (enter amount when prompted)
- `quit`: Exit the market

Demo Sequence

1. Show multiple buyers connecting:

- Start 4+ buyer windows
- Each should show connection confirmation

2. Demonstrate listing:

- Type `list` in any buyer window
- Shows current item, amount, and time left

3. Demonstrate purchases:

- Type `buy` in different buyer windows
- Enter amount to purchase
- All buyers receive stock updates

4. Demonstrate item switching:

- Either wait 60 seconds
- Or buy all stock of current item
- All buyers receive new item notification

5. Show buyer disconnection:

- Type `quit` in any buyer window

- Buyer exits cleanly

Note

- Seller must be started before any buyers
- Each buyer needs a unique ID
- All windows should be kept open during demo